# DOMiNANCE CVS MOCK EXAMINATION 01

**Cardiovascular Physiology | 60 questions | 90 minutes | 60 marks**

Choose the single best answer (A-D) for each question. The answer key is on the page immediately after the last question.

**1. A deep vein in one leg becomes obstructed, increasing pressure in vessels upstream. What local change most directly favours swelling?**

A. Raised capillary hydrostatic pressure
B. A decrease in capillary hydrostatic pressure
C. An increase in plasma albumin concentration
D. An increase in lymphatic drainage capacity

**2. After firm stroking of the skin, redness spreads beyond the directly stimulated line. What produces this surrounding flare?**

A. Closure of all precapillary sphincters around the area
B. Direct filtration of red blood cells into the epidermis
C. Clotting of capillaries at the site of injury
D. An axon reflex causing nearby arterioles to dilate

**3. In a healthy adult, cuff pressure is slowly released during blood-pressure measurement. Disappearance of Korotkoff sounds usually marks which pressure?**

A. Mean arterial pressure
B. Pulse pressure
C. Systolic pressure in the brachial artery
D. Diastolic arterial pressure

**4. A patient develops rapidly rising intracranial pressure with impaired brainstem perfusion. Which combination is typical of the Cushing response?**

A. Increased arterial pressure with reflex slowing of the heart
B. Decreased arterial pressure with complete loss of sympathetic tone
C. Increased arterial pressure with sustained reflex tachycardia
D. Decreased arterial pressure with reflex slowing of the heart

**5. Which structure forms the key physical restriction to paracellular movement across the blood-brain barrier?**

A. Endothelial tight junctions.
B. The choroid plexus epithelium.
C. Loose junctions of cerebral venules.
D. Astrocytic end-feet alone.

**6. Why does the sinoatrial (SA) node normally determine the heart rate?**

A. Its impulses reach ventricular myocardium faster than those in Purkinje fibres.
B. It reaches the threshold for spontaneous discharge sooner than other pacemakers.
C. Its cells have a more negative resting membrane potential than ventricular cells.
D. It produces the largest action potential of all pacemaker cells.

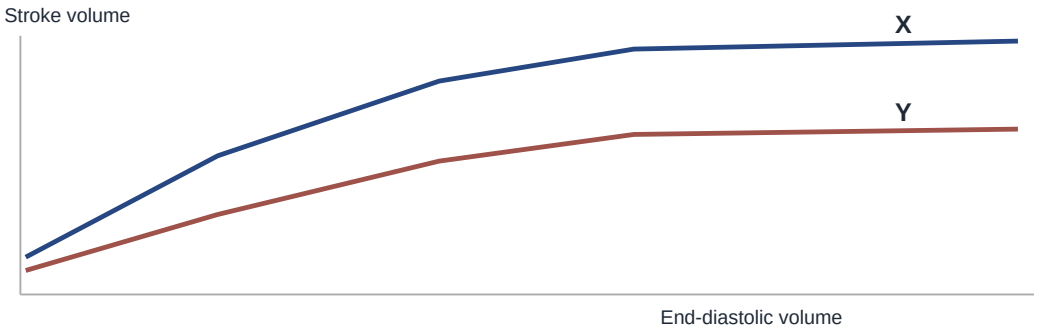
**7. Two beats have different QRS heights. A student concludes the taller complex produced a larger stroke volume. What is the main flaw?**

A. QRS height measures the amount of venous blood returning.
B. QRS height represents the duration of ventricular contraction.
C. QRS height directly records the pressure within the ventricle.
D. QRS height reflects voltage, not the amount of blood ejected.

**8. Why does most left ventricular coronary blood flow occur during diastole rather than systole?**

A. Pulmonary venous pressure becomes higher than aortic pressure during systole.
B. Contracting left ventricular muscle compresses vessels within its wall.
C. The left coronary artery is closed by the aortic valve during systole.
D. The heart stops using oxygen during systole.

**9. Two ventricular function curves are shown. At the same end-diastolic volume, curve X produces a larger stroke volume than curve Y. What best explains the difference?**




A. Ventricular preload must be greater on curve X.
B. The ventricular muscle must be less responsive to calcium on curve X.
C. Ventricular contractility is greater on curve X.
D. The heart rate must be lower on curve X.

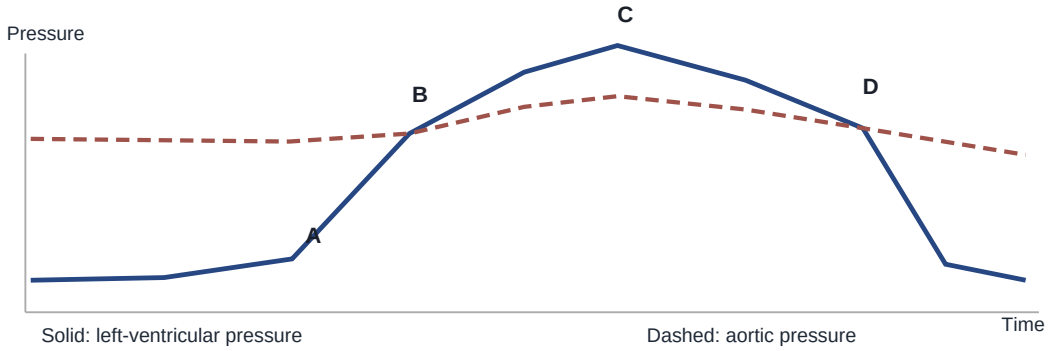
**10. Inflation of the lungs sends sensory impulses towards the medulla. What cardiovascular response is described in the lecture?**

A. Vagal afferent inhibition of vasomotor activity with vasodilation.
B. Carotid sinus chemoreceptor inhibition with increased arterial tone.
C. Atrial volume-receptor excitation with sustained vasoconstriction.
D. Sympathetic afferent excitation of vasomotor activity with constriction.

**11. What change occurs in the pulmonary circulation with the newborn's first breaths?**

A. The ductus venosus begins carrying blood from the right ventricle.
B. Pulmonary vascular resistance rises as alveolar oxygen rises.
C. Systemic vascular resistance falls because the placenta remains connected.
D. Pulmonary vascular resistance falls as the lungs expand and oxygen rises.

**12. In the schematic left-ventricular and aortic pressure tracings, which labelled point marks the initial opening of the aortic valve?**




A. Point B
B. Point C
C. Point A
D. Point D

**13. After bleeding stops, both haemoglobin and plasma-protein concentrations fall as circulating volume recovers. What explains this?**

A. New red-cell production lowers haemoglobin concentration.
B. Venoconstriction selectively removes haemoglobin from blood.
C. Added fluid dilutes the remaining blood components.
D. Raised renal filtration removes most circulating proteins.

**14. The cerebral aqueduct becomes blocked while CSF production continues. Which ventricles are expected to enlarge first?**

A. The fourth ventricle alone
B. The third and fourth ventricles only
C. The lateral and third ventricles
D. The fourth ventricle and central canal only

**15. What prevents ventricular myocytes from sustaining a normal tetanus?**

A. Systolic calcium release becomes too small to sustain successive twitches.
B. Spontaneous pacemaker activity prevents ventricular re-excitation.
C. Ventricular muscle has a refractory period shorter than its twitch.
D. Refractoriness lasts through most of contraction.

**16. In a lead where the QRS is upright, the T wave is often upright too. Which combined explanation best accounts for this?**

A. Repolarisation has the same polarity but travels in the reverse direction.
B. Repolarisation has opposite polarity and generally travels in the reverse direction.
C. Repolarisation has the same electrical sign and travels in the same direction through the ventricles.
D. Repolarisation has opposite polarity but travels in the same direction.

**17. Mean arterial pressure is 95 mmHg and intracranial pressure is 15 mmHg. Assuming cerebral venous pressure is not higher than intracranial pressure, what is cerebral perfusion pressure?**

A. 65 mmHg
B. 95 mmHg
C. 75 mmHg
D. 80 mmHg

**18. Left ventricular pressure has risen to 45 mmHg. Left atrial pressure is 10 mmHg and aortic pressure is 85 mmHg. Which phase is most likely?**

A. Isovolumetric contraction
B. Isovolumetric relaxation
C. Reduced ventricular ejection
D. Rapid ventricular filling

**19. Blood flow through a forearm artery is briefly blocked and then restored. Why may flow temporarily rise above its resting level?**

A. Venous outflow rises suddenly and forces extra blood through the artery.
B. Restoration of pressure directly relaxes arteriolar smooth muscle.
C. The release of occlusion briefly raises systemic arterial pressure.
D. Accumulated metabolites continue to dilate local vessels.

**20. Why can prolonged renin-angiotensin-aldosterone activation worsen heart-failure oedema?**

A. It causes sustained urinary sodium loss.
B. It decreases circulating volume and filling pressure.
C. Salt and water retention raises venous pressure.
D. It prevents compensatory arterial vasoconstriction.

**21. According to Poiseuille's law, if a vessel radius is reduced to half while other factors remain constant, how does resistance change?**

A. It becomes 4 times greater.
B. It becomes 16 times greater.
C. It becomes roughly two times greater.
D. It becomes 8 times greater.

**22. Damage selectively removes the inhibitory effect of the medullary depressor area on the pressor area. Which response is expected?**

A. Reduced sympathetic discharge with markedly decreased vascular tone.
B. Greater vagal discharge with isolated venous dilation.
C. Greater sympathetic discharge with increased vascular tone.
D. Reduced local metabolism with fixed arteriolar diameter.

**23. A person loses a substantial amount of blood. What sequence most directly explains the initial fall in cardiac output?**

A. Lower blood volume directly raises ventricular end-diastolic volume.
B. Lower blood volume raises pulmonary venous return above systemic venous return.
C. Lower blood volume immediately increases plasma oncotic pressure enough to raise output.
D. Lower blood volume leads to lower venous return and ventricular filling.

**24. A drug markedly inhibits SERCA in cardiac muscle. What change in relaxation is expected?**

A. Slower calcium removal impairs relaxation.
B. Its calcium influx through L-type channels immediately increases.
C. Shorter calcium transient and stronger relaxation.
D. Faster removal of calcium from cytosol.

**25. At 25 mm/s, a PR interval extends across 4.5 small squares. Which interpretation is most accurate?**

A. 0.24 s, above the stated PR-interval range.
B. 0.90 s, above the stated PR-interval range.
C. 0.09 s, shorter than the stated normal PR-interval range.
D. 0.18 s, inside the stated PR-interval range.

**26. A patient develops headache that is worse upright after lumbar CSF sampling. Which mechanism is most likely?**

A. The procedure obstructs both foramina of Monro.
B. Persistent CSF leakage lowers intracranial CSF pressure.
C. Lumbar puncture increases CSF production several-fold.
D. The sample immediately increases blood-brain barrier permeability.

**27. Why does inspiration normally widen the split between the aortic and pulmonary components of the second heart sound?**

A. More right-heart filling delays pulmonary valve closure.
B. Reduced right ventricular filling delays aortic valve closure.
C. Increased right ventricular filling delays mitral valve opening.
D. Increased left ventricular filling delays tricuspid valve closure.

**28. A patient with left ventricular failure develops breathlessness and pulmonary oedema. What pressure change most directly promotes fluid movement into the lungs?**

A. Increased lymphatic drainage with unchanged capillary pressure
B. Increased pulmonary venous and capillary hydrostatic pressure
C. Increased plasma oncotic pressure in pulmonary capillaries
D. Decreased hydrostatic pressure in pulmonary capillaries

**29. Rapid infusion of fluid increases venous return and stretches the right atrium. Which reflex response may help accommodate the larger venous return?**

A. Heart rate may remain unchanged because atrial stretch has no reflex effects.
B. Heart rate may increase as atrial stretch activates a reflex.
C. Heart rate rises because the fluid infusion has reduced carotid sinus pressure.
D. Heart rate may decrease because atrial stretch directly increases vagal outflow.

**30. Aortic pressure suddenly rises. For the first affected beat, ventricular filling and contractility are unchanged. What is the likely effect on stroke volume?**

A. Stroke volume increases because ventricular filling has increased.
B. Stroke volume increases because aortic recoil becomes stronger.
C. Stroke volume stays constant because heart rate has not changed.
D. Stroke volume decreases with greater aortic afterload.

**31. Why can a cerebral arterial branch occlusion cause a local infarct despite the Circle of Willis?**

A. Small distal territories may have limited effective collateral flow.
B. The basilar artery can supply every distal cortical branch directly.
C. Every cerebral artery has a complete set of parallel branches.
D. Cerebral veins normally replace arterial inflow after blockage.

**32. An ECG develops a new shift of the ST segment above or below the baseline. Which process specifically makes this segment clinically important?**

A. Normal atrial depolarisation creates the ST deflection of each beat.
B. The onset of valve closure normally lifts the ST segment above baseline.
C. A decrease in cardiac preload directly determines ST segment height.
D. Myocardial ischaemia or inadequate oxygenation can alter ST level.

**33. Oxygen consumption is 280 mL/min. Arterial oxygen content is 200 mL/L and mixed venous oxygen content is 130 mL/L. What is cardiac output?**

A. 5 L/min
B. 2 L/min
C. 4 L/min
D. 3 L/min

**34. Which observation most strongly points to a myogenic rather than metabolic adjustment?**

A. Dilation follows reduced tissue oxygen supply.
B. Dilation follows accumulation of local lactate.
C. Constriction follows a rise in vessel-wall stretch.
D. Constriction follows removal of excess local metabolites.

**35. Why is the normal T wave generally broader than the QRS complex?**

A. The T wave corresponds to ventricular contraction rather than electricity.
B. Ventricular repolarisation usually takes longer than its depolarisation.
C. The T wave covers atrial and ventricular depolarisation together.
D. Ventricular depolarisation spreads more slowly than repolarisation.

**36. A resting adult has a blood pressure of 120/75 mmHg. Using the usual resting approximation, what is mean arterial pressure?**

A. 95 mmHg
B. 75 mmHg
C. 85 mmHg
D. 90 mmHg

**37. Portal hypertension promotes ascites partly by increasing which force?**

A. Hydrostatic pressure in hepatic sinusoids.
B. Interstitial lymphatic pumping efficiency.
C. Arterial oxygen tension in the hepatic vein.
D. Plasma oncotic pressure within hepatic capillaries.

**38. An older patient faints shortly after standing to urinate at night. Which phenomenon does this episode most closely illustrate?**

A. Carotid sinus syncope caused by neck pressure.
B. Septic shock due to widespread inflammatory vasodilatation.
C. Micturition syncope linked to reflex hypotension.
D. Positive-g blackout from head-level underperfusion.

**39. Why might severe anaemia produce a cardiac flow murmur despite structurally normal valves?**

A. Anaemia markedly increases blood viscosity and suppresses flow.
B. Lower viscosity and increased flow favour turbulence.
C. Anaemia blocks the ventricular conduction system.
D. Reduced blood volume necessarily closes the aortic valve.

**40. At 25 mm/s, consecutive ECG R waves are separated by 20 small horizontal squares. What is the ventricular rate?**

A. 100 beats/min
B. 75 beats/min
C. 60 beats/min
D. 80 beats/min

**41. Immediately after a person stands up quickly, before reflex compensation has fully developed, what tends to happen?**

A. Venous pooling decreases and ventricular filling rises.
B. Cardiac output rises because gravity pushes blood toward the chest.
C. Venous pooling increases and venous return to the heart falls.
D. Venous pressure in the feet falls below pressure at heart level.

**42. How can an impulse spread rapidly between adjacent cardiac muscle cells?**

A. Gap junctions provide low-resistance electrical connections.
B. Each cell must first receive a motor-neuron endplate signal.
C. Mechanical desmosomes independently generate pacemaker signals.
D. The connective tissue between cells conducts sodium current.

**43. Why does fetal haemoglobin load oxygen effectively at the placental interface?**

A. It binds 2,3-BPG more tightly than adult haemoglobin.
B. Its weaker 2,3-BPG binding increases oxygen affinity.
C. Its alpha-globin chains bind more oxygen than maternal alpha chains.
D. Each fetal haemoglobin molecule carries more oxygen binding sites than adult haemoglobin.

**44. An organ's flow remains nearly constant as its perfusion pressure rises across the autoregulatory range. Which vascular adjustment produces this plateau?**

A. Arteriolar resistance falls as perfusion pressure rises.
B. Venous blood begins flowing backward into the artery.
C. Cardiac output stops changing throughout the entire body.
D. Arteriolar resistance rises as perfusion pressure rises.

**45. A ventricle has an end-diastolic volume of 140 mL and an end-systolic volume of 56 mL. What is its ejection fraction?**

A. 84%
B. 65%
C. 50%
D. 60%

**46. During exercise, pulmonary blood flow increases greatly but pulmonary arterial pressure rises relatively little. What helps explain this?**

A. Pulmonary vascular resistance increases in direct proportion to flow.
B. The alveoli stop exchanging gases while blood flow rises.
C. Previously underperfused capillaries open and existing vessels widen.
D. Most pulmonary capillaries close as flow increases.

**47. A patient with severe infection has widespread arteriolar dilation and low arterial pressure. Which shock mechanism is most likely?**

A. Distributive shock from widespread loss of vascular tone
B. Cardiogenic shock caused mainly by poor ventricular pumping
C. Obstructive shock caused mainly by mechanical blockage of blood flow
D. Hypovolaemic shock caused mainly by external blood loss

**48. Why is average blood-flow velocity normally much lower in capillaries than in the aorta?**

A. Capillaries receive a smaller total blood flow than the aorta in steady state.
B. The total capillary cross-sectional area is smaller than that of the aorta.
C. The combined cross-sectional area of all capillaries is much larger.
D. Lower capillary pressure alone explains the slower movement of red cells.

**49. A patient has low arterial oxygen and high carbon dioxide. Which peripheral sensors respond mainly to these chemical changes?**

A. The atrial type A and type B volume receptors.
B. The carotid bodies and the aortic bodies.
C. The carotid sinus and aortic arch baroreceptors.
D. The sinoatrial and atrioventricular pacemaker nodes.

**50. Which feature most reliably distinguishes first-degree AV block from second-degree AV block in the chapter's basic ECG classification?**

A. Both have identical PR intervals on each beat.
B. First-degree conducts every P; second-degree drops some.
C. First-degree lacks P waves; second-degree lacks QRS complexes.
D. Both show independent P-wave and QRS rhythms.

**51. At the same stroke volume and heart rate, which change generally increases myocardial oxygen demand more?**

A. Pumping against a much higher arterial pressure
B. Lowering wall tension during ventricular ejection
C. Pumping against a lower arterial pressure
D. Reducing the pressure needed to open the aortic valve

**52. Immediately after arterial pressure falls during haemorrhage, which autonomic response follows reduced arterial baroreceptor firing?**

A. Reduced sympathetic and reduced vagal activity together
B. Increased vagal activity with arteriolar dilation
C. Increased sympathetic activity and reduced vagal activity
D. Reduced sympathetic activity and increased vagal activity

**53. A patient has near-normal resting cardiac output but very high ventricular filling pressures and pulmonary congestion. Which statement fits best?**

A. Heart failure can occur with normal output.
B. High filling pressure proves adequate pumping.
C. Normal resting output excludes heart failure.
D. Heart failure requires reduced ejection fraction.

**54. A patient develops chronic swelling of an arm after axillary lymph nodes are removed. What is the main reason?**

A. Plasma albumin production increases sharply in the affected tissue.
B. The local arterial inflow rises more than the venous outflow can accommodate.
C. Protein-rich tissue fluid cannot drain through the lymphatics.
D. Venous pressure is elevated, increasing filtration through intact capillaries.

**55. Why is aVF classed as an augmented unipolar limb lead?**

A. It compares two fixed chest electrodes in the horizontal plane of the heart.
B. It uses the left-foot electrode against an averaged limb reference.
C. It compares the right arm directly against the left arm.
D. It uses the right-leg electrode as its positive terminal.

**56. One group of alveoli is poorly ventilated while nearby alveoli are well ventilated. What happens to the small pulmonary arteries supplying the poorly ventilated region?**

A. They dilate through the same low-oxygen response as systemic arterioles.
B. They dilate, directing more blood toward the poorly ventilated alveoli.
C. They constrict, directing more blood toward better-ventilated alveoli.
D. They constrict in response to reduced systemic pressure rather than local oxygen.

**57. Which isolated change increases the tendency for turbulent flow in a blood vessel?**

A. A substantial rise in whole-blood viscosity
B. An increase in blood-flow velocity
C. A decrease in blood density
D. A decrease in vessel diameter

**58. Venous return rises while heart rate, afterload and contractility remain unchanged. What immediate change is expected in a normal ventricle?**

A. A rise in stroke volume with a fall in end-diastolic volume
B. A fall in stroke volume despite a rise in end-diastolic volume
C. A fall in end-diastolic volume and stroke volume
D. A rise in end-diastolic volume and stroke volume

**59. What distinguishes the normal x descent from the y descent in a JVP trace?**

A. x: atrial relaxation; y: emptying through the open tricuspid.
B. x: aortic valve opening; y: mitral valve closure.
C. x: atrial contraction; y: ventricular isovolumetric contraction.
D. x: ventricular ejection; y: pulmonary arterial recoil.

**60. In complete AV dissociation, an atrium sometimes contracts while the tricuspid valve is closed. Which jugular venous finding may appear?**

A. Loss of the y descent with every normal beat
B. A persistently absent a wave
C. Intermittent giant a waves
D. Intermittent giant v waves caused by atrial systole

# ANSWER KEY

Mock 01 - 60 Questions | One mark each

Check your answers only after completing the examination.

| Q01-Q15 | | Q16-Q30 | | Q31-Q45 | | Q46-Q60 | |
|---|---|---|---|---|---|---|---|
| 01. | A | 16. | B | 31. | A | 46. | C |
| 02. | D | 17. | D | 32. | D | 47. | A |
| 03. | D | 18. | A | 33. | C | 48. | C |
| 04. | A | 19. | D | 34. | C | 49. | B |
| 05. | A | 20. | C | 35. | B | 50. | B |
| 06. | B | 21. | B | 36. | D | 51. | A |
| 07. | D | 22. | C | 37. | A | 52. | C |
| 08. | B | 23. | D | 38. | C | 53. | A |
| 09. | C | 24. | A | 39. | B | 54. | C |
| 10. | A | 25. | D | 40. | B | 55. | B |
| 11. | D | 26. | B | 41. | C | 56. | C |
| 12. | A | 27. | A | 42. | A | 57. | B |
| 13. | C | 28. | B | 43. | B | 58. | D |
| 14. | C | 29. | B | 44. | D | 59. | A |
| 15. | D | 30. | D | 45. | D | 60. | C |

Answer key ends here. Review explanations separately in the question-bank archive.